

2014 January Q5 C34

5. (a) Prove, by using logarithms, that

$$\frac{d}{dx}(2^x) = 2^x \ln 2$$

(3)

The curve C has the equation

$$2x + 3y^2 + 3x^2y + 12 = 4 \times 2^x$$

The point P , with coordinates $(2, 0)$, lies on C .

- (b) Find an equation of the tangent to C at P .

(6)

2014 June Q2 C34

2. A curve C has the equation

$$x^3 - 3xy - x + y^3 - 11 = 0$$

Find an equation of the tangent to C at the point $(2, -1)$, giving your answer in the form $ax + by + c = 0$, where a , b and c are integers.

(6)

2014 June Q8 C34

8. The volume V of a spherical balloon is increasing at a constant rate of $250 \text{ cm}^3 \text{ s}^{-1}$. Find the rate of increase of the radius of the balloon, in cm s^{-1} , at the instant when the volume of the balloon is $12\,000 \text{ cm}^3$. Give your answer to 2 significant figures.

(5)

[You may assume that the volume V of a sphere of radius r is given by the formula $V = \frac{4}{3}\pi r^3$.]

2015 January Q6 C34

6. (i) Given $x = \tan^2 4y$, $0 < y < \frac{\pi}{8}$, find $\frac{dy}{dx}$ as a function of x .

Write your answer in the form $\frac{1}{A(x^p + x^q)}$, where A , p and q are constants to be found.

(5)

- (ii) The volume V of a cube is increasing at a constant rate of $2 \text{ cm}^3 \text{ s}^{-1}$. Find the rate at which the length of the edge of the cube is increasing when the volume of the cube is 64 cm^3 .

(5)

2015 June Q1 C34

1. A curve has equation

$$4x^2 - y^2 + 2xy + 5 = 0$$

The points P and Q lie on the curve.

Given that $\frac{dy}{dx} = 2$ at P and at Q ,

- (a) use implicit differentiation to show that $y - 6x = 0$ at P and at Q .

(6)

- (b) Hence find the coordinates of P and Q .

(3)

10.

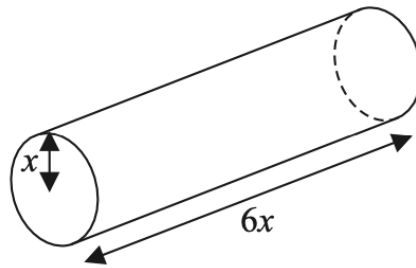


Figure 4

Figure 4 shows a right circular cylindrical rod which is expanding as it is heated.

At time t seconds the radius of the rod is x cm and the length of the rod is $6x$ cm.

Given that the **cross-sectional area** of the rod is increasing at a constant rate of $\frac{\pi}{20} \text{ cm}^2 \text{ s}^{-1}$, find the rate of increase of the volume of the rod when $x = 2$

Write your answer in the form $k\pi \text{ cm}^3 \text{ s}^{-1}$ where k is a rational number.

(6)

2016 January Q3 C34

3. A curve C has equation

$$3^x + 6y = \frac{3}{2}xy^2$$

Find the exact value of $\frac{dy}{dx}$ at the point on C with coordinates $(2, 3)$. Give your answer in the form $\frac{a + \ln b}{8}$, where a and b are integers.

(7)

2016 January Q13 C34

13. A curve C has parametric equations

$$x = 6\cos 2t, \quad y = 2\sin t, \quad -\frac{\pi}{2} < t < \frac{\pi}{2}$$

(a) Show that $\frac{dy}{dx} = \lambda \operatorname{cosec} t$, giving the exact value of the constant λ .

(4)

(b) Find an equation of the normal to C at the point where $t = \frac{\pi}{3}$

Give your answer in the form $y = mx + c$, where m and c are simplified surds.

(6)

The cartesian equation for the curve C can be written in the form

$$x = f(y), \quad -k < y < k$$

where $f(y)$ is a polynomial in y and k is a constant.

(c) Find $f(y)$.

(3)

(d) State the value of k .

(1)

2016 June Q2 C34

2. The point P with coordinates $\left(\frac{\pi}{2}, 1\right)$ lies on the curve with equation

$$4x \sin x = \pi y^2 + 2x, \quad \frac{\pi}{6} \leq x \leq \frac{5\pi}{6}$$

Find an equation of the normal to the curve at P .

(6)

13.

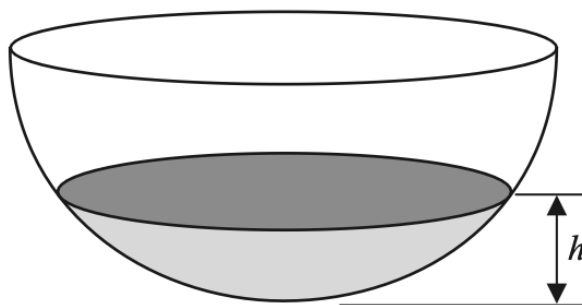


Figure 4

Figure 4 shows a hemispherical bowl containing some water.

At t seconds, the height of the water is h cm and the volume of the water is V cm³, where

$$V = \frac{1}{3}\pi h^2(30 - h), \quad 0 < h \leq 10$$

The water is leaking from a hole in the bottom of the bowl.

Given that $\frac{dV}{dt} = -\frac{1}{10}V$

(a) show that $\frac{dh}{dt} = -\frac{h(30 - h)}{30(20 - h)}$ (5)

(b) Write $\frac{30(20 - h)}{h(30 - h)}$ in partial fraction form. (3)

Given that $h = 10$ when $t = 0$,

(c) use your answers to parts (a) and (b) to find the time taken for the height of the water to fall to 5 cm. Give your answer in seconds to 2 decimal places. (6)

2017 January Q1 C34

1. Find an equation of the tangent to the curve

$$x^3 + 3x^2y + y^3 = 37$$

at the point $(1, 3)$. Give your answer in the form $ax + by + c = 0$, where a , b and c are integers.

(6)

13.

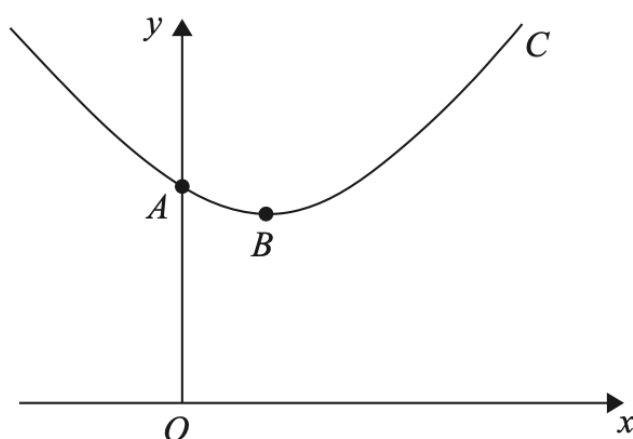


Figure 4

The curve C shown in Figure 4 has parametric equations

$$x = 1 + \sqrt{3} \tan \theta, \quad y = 5 \sec \theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

The curve C crosses the y -axis at A and has a minimum turning point at B , as shown in Figure 4.

(a) Find the exact coordinates of A . (3)

(b) Show that $\frac{dy}{dx} = \lambda \sin \theta$, giving the exact value of the constant λ . (4)

(c) Find the coordinates of B . (2)

(d) Show that the cartesian equation for the curve C can be written in the form

$$y = k\sqrt{(x^2 - 2x + 4)}$$

where k is a simplified surd to be found.

(3)

2017 June Q1 C34

1. A curve C has equation

$$3x^2 + 2xy - 2y^2 + 4 = 0$$

Find an equation for the tangent to C at the point $(2, 4)$, giving your answer in the form $ax + by + c = 0$ where a , b and c are integers.

(6)

12.

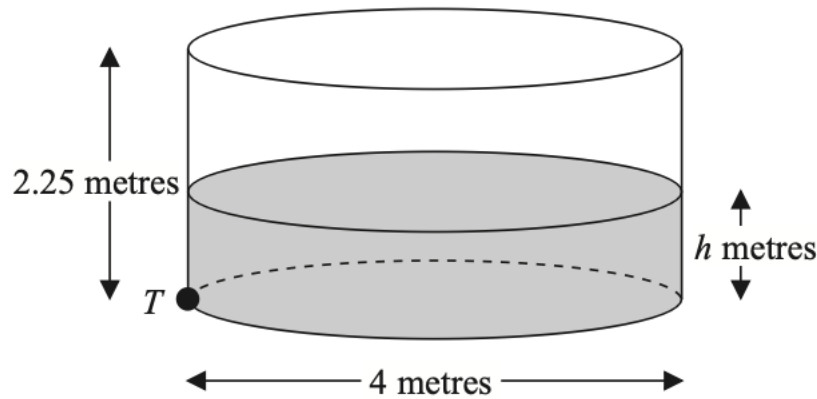


Figure 4

Figure 4 shows a right cylindrical water tank. The diameter of the circular cross section of the tank is 4 m and the height is 2.25 m. Water is flowing into the tank at a constant rate of $0.4\pi \text{ m}^3 \text{ min}^{-1}$. There is a tap at a point T at the base of the tank. When the tap is open, water leaves the tank at a rate of $0.2\pi\sqrt{h} \text{ m}^3 \text{ min}^{-1}$, where h is the height of the water in metres.

- (a) Show that at time t minutes after the tap has been opened, the height h m of the water in the tank satisfies the differential equation

$$20 \frac{dh}{dt} = 2 - \sqrt{h} \quad (5)$$

At the instant when the tap is opened, $t = 0$ and $h = 0.16$

- (b) Use the differential equation to show that the time taken to fill the tank to a height of 2.25 m is given by

$$\int_{0.16}^{2.25} \frac{20}{2 - \sqrt{h}} dh \quad (2)$$

Using the substitution $h = (2 - x)^2$, or otherwise,

- (c) find the time taken to fill the tank to a height of 2.25 m.

Give your answer in minutes to the nearest minute.

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(7)

2017 October Q2 C34

2. The curve C has equation

$$y^3 + x^2y - 6x = 0$$

(a) Find $\frac{dy}{dx}$ in terms of x and y .

(5)

(b) Hence find the exact coordinates of the points on C for which $\frac{dy}{dx} = 0$

(6)

10.

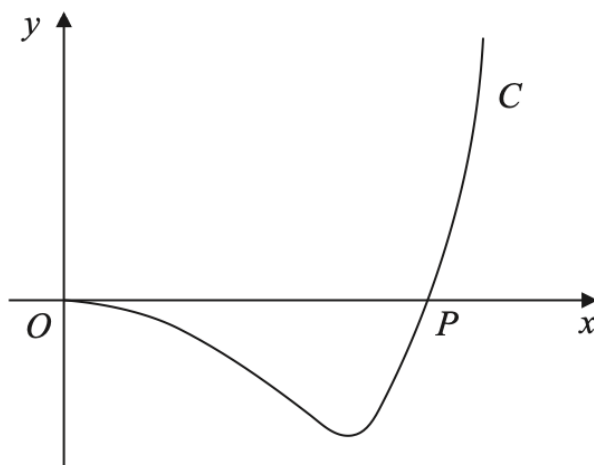


Figure 2

Figure 2 shows a sketch of part of the curve C with parametric equations

$$x = \frac{20t}{2t+1} \quad y = t(t-4), \quad t > 0$$

The curve cuts the x -axis at the point P .

(a) Find the x coordinate of P .

(2)

(b) Show that $\frac{dy}{dx} = \frac{(t-A)(2t+1)^2}{B}$ where A and B are constants to be found.

(5)

(c) (i) Make t the subject of the formula

$$x = \frac{20t}{2t+1}$$

(ii) Hence find a cartesian equation of the curve C . Write your answer in the form

$$y = f(x), \quad 0 < x < k$$

where $f(x)$ is a single fraction and k is a constant to be found.

(6)

2018 January Q1 C34

1. A curve C has equation

$$3^x + xy = x + y^2, \quad y > 1$$

The point P with coordinates $(4, 11)$ lies on C .

Find the exact value of $\frac{dy}{dx}$ at the point P .

Give your answer in the form $a + b \ln 3$, where a and b are rational numbers.

(6)

2018 January Q8 C34

8. Given that

$$y = 8 \tan(2x), \quad -\frac{\pi}{4} < x < \frac{\pi}{4}$$

show that

$$\frac{dx}{dy} = \frac{A}{B + y^2}$$

where A and B are integers to be found.

(4)

7.

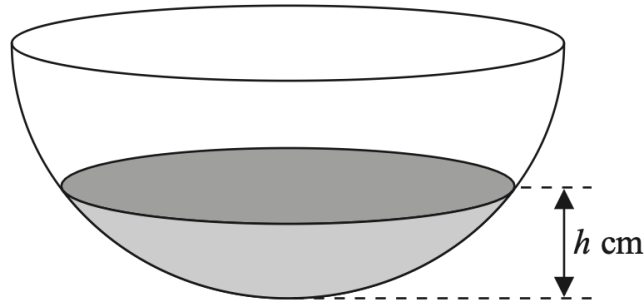


Figure 1

Figure 1 shows a hemispherical bowl.

Water is flowing into the bowl at a constant rate of $180 \text{ cm}^3 \text{ s}^{-1}$.

When the height of the water is $h \text{ cm}$, the volume of water $V \text{ cm}^3$ is given by

$$V = \frac{1}{3} \pi h^2 (90 - h), \quad 0 \leq h \leq 30$$

Find the rate of change of the height of the water, in cm s^{-1} , when $h = 15$

Give your answer to 2 significant figures.

(5)

2018 June Q10 C34

10. The curve C satisfies the equation

$$xe^{5-2y} - y = 0 \quad x > 0, \ y > 0$$

The point P with coordinates $(2e^{-1}, 2)$ lies on C .

The tangent to C at P cuts the x -axis at the point A and cuts the y -axis at the point B .

Given that O is the origin, find the exact area of triangle OAB , giving your answer in its simplest form.

(7)

2018 October Q2 C34

2. A curve C has equation

$$x^3 - 4xy + 2x + 3y^2 - 3 = 0$$

Find an equation of the normal to C at the point $(-3, 2)$, giving your answer in the form $ax + by + c = 0$ where a , b and c are integers.

(7)

2018 October Q13 C34

13. The volume of a spherical balloon of radius r m is V m³, where $V = \frac{4}{3}\pi r^3$

(a) Find $\frac{dV}{dr}$ (1)

Given that the volume of the balloon increases with time t seconds according to the formula

$$\frac{dV}{dt} = \frac{20}{V(0.05t + 1)^3}, \quad t \geq 0$$

(b) find an expression in terms of r and t for $\frac{dr}{dt}$ (3)

Given that $V = 1$ when $t = 0$

(c) solve the differential equation

$$\frac{dV}{dt} = \frac{20}{V(0.05t + 1)^3}$$

giving your answer in the form $V^2 = f(t)$. (6)

(d) Hence find the radius of the balloon at time $t = 20$, giving your answer to 3 significant figures. (3)

2019 January Q4 C34

4. The curve C has equation

$$81y^3 + 64x^2y + 256x = 0$$

(a) Find $\frac{dy}{dx}$ in terms of x and y .

(5)

(b) Hence find the coordinates of the points on C where $\frac{dy}{dx} = 0$

(6)

10.

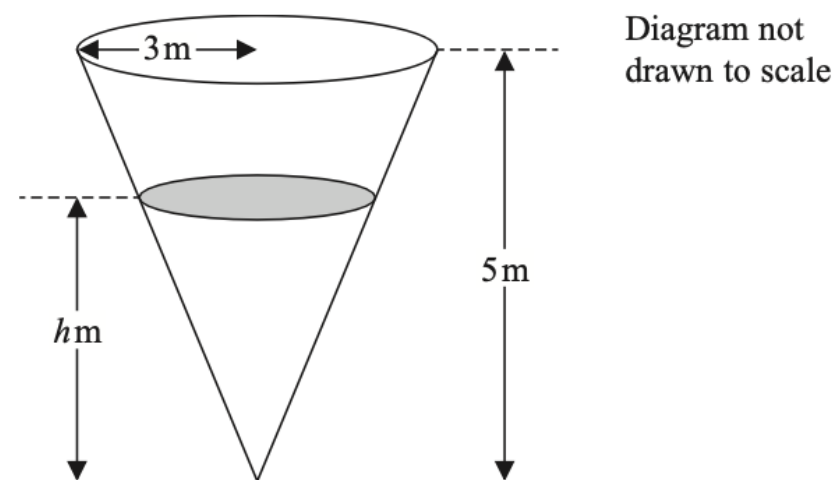


Figure 3

Figure 3 shows a container in the shape of an inverted right circular cone which contains some water.

The cone has an internal radius of 3 m and a vertical height of 5 m as shown in Figure 3.

At time t seconds, the height of the water is h metres, the volume of the water is $V\text{ m}^3$ and water is leaking from a hole in the bottom of the container at a constant rate of $0.02\text{ m}^3\text{ s}^{-1}$

[The volume of a cone of radius r and height h is $\frac{1}{3}\pi r^2 h$.]

(a) Show that, while the water is leaking,

$$h^2 \frac{dh}{dt} = -\frac{1}{k\pi}$$

where k is a constant to be found.

(5)

Given that the container is initially full of water,

(b) express h in terms of t .

(3)

(c) Find the time taken for the container to empty, giving your answer to the nearest minute.

(2)

2019 June Q4 C34

4. The curve C has equation

$$3ye^{-2x} = 4x^2 + y^2 + 2$$

- (a) Find $\frac{dy}{dx}$ in terms of x and y .

(5)

The point P on C has coordinates $(0, 2)$.

- (b) Find the equation of the normal to C at P giving your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

2020 October Q6 P4

6. A curve C has equation

$$y = x^{\sin x} \quad x > 0 \quad y > 0$$

- (a) Find, by firstly taking natural logarithms, an expression for $\frac{dy}{dx}$ in terms of x and y . (5)
- (b) Hence show that the x coordinates of the stationary points of C are solutions of the equation

$$\tan x + x \ln x = 0$$

(2)

2021 January Q6 P4

6. A curve has equation

$$4y^2 + 3x = 6ye^{-2x}$$

(a) Find $\frac{dy}{dx}$ in terms of x and y .

(5)

The curve crosses the y -axis at the origin and at the point P .

(b) Find the equation of the normal to the curve at P , writing your answer in the form $y = mx + c$ where m and c are constants to be found.

(4)

3.

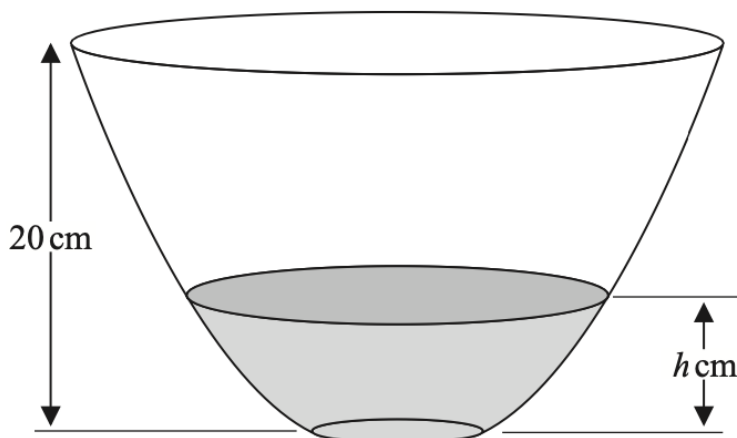


Figure 2

A bowl with circular cross section and height 20 cm is shown in Figure 2.

The bowl is initially empty and water starts flowing into the bowl.

When the depth of water is h cm, the volume of water in the bowl, $V \text{ cm}^3$, is modelled by the equation

$$V = \frac{1}{3}h^2(h + 4) \quad 0 \leq h \leq 20$$

Given that the water flows into the bowl at a constant rate of $160 \text{ cm}^3 \text{ s}^{-1}$, find, according to the model,

(a) the time taken to fill the bowl, (2)

(b) the rate of change of the depth of the water, in cm s^{-1} , when $h = 5$ (5)

2021 June Q5 P4

5. A curve has equation

$$y^2 = ye^{-2x} - 3x$$

(a) Show that

$$\frac{dy}{dx} = \frac{2ye^{-2x} + 3}{e^{-2x} - 2y} \quad (4)$$

The curve crosses the y -axis at the origin and at the point P .

The tangent to the curve at the origin and the tangent to the curve at P meet at the point R .

(b) Find the coordinates of R .

(5)

2021 October Q1 P4

1. The curve C has equation

$$2x - 4y^2 + 3x^2y = 4x^2 + 8$$

The point $P(3, 2)$ lies on C .

Find the equation of the normal to C at the point P , writing your answer in the form $ax + by + c = 0$ where a , b and c are integers to be found.

(7)

9.

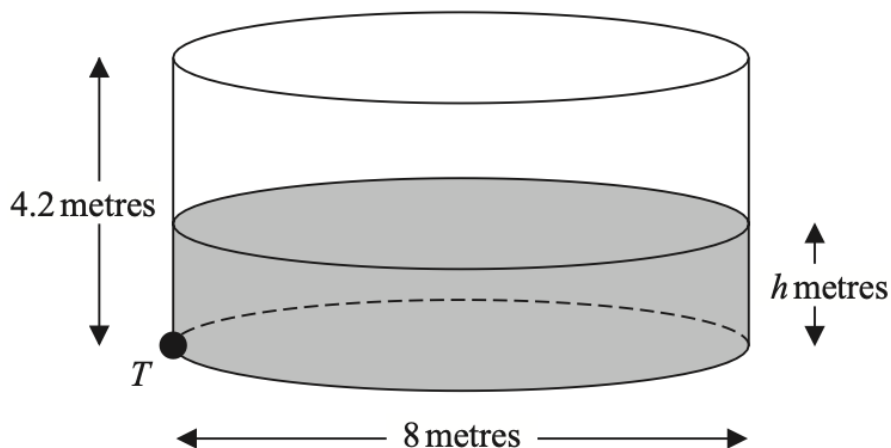


Figure 4

Figure 4 shows a cylindrical tank that contains some water.

The tank has an internal diameter of 8 m and an internal height of 4.2 m.

Water is flowing into the tank at a constant rate of $(0.6\pi)\text{ m}^3$ per minute.

There is a tap at point T at the bottom of the tank.

At time t minutes after the tap has been opened,

- the depth of the water is h metres
- the water is leaving the tank at a rate of $(0.15\pi h)\text{ m}^3$ per minute

(a) Show that

$$\frac{dh}{dt} = \frac{12 - 3h}{320}$$

(4)

Given that the depth of the water in the tank is 0.5 m when the tap is opened,

(b) find the time taken for the depth of water in the tank to reach 3.5 m.

(6)

2022 January Q1 P4

1. The curve C has equation

$$xy^2 = x^2y + 6 \quad x \neq 0 \quad y \neq 0$$

Find an equation for the tangent to C at the point $P(2, 3)$, giving your answer in the form $ax + by + c = 0$ where a , b and c are integers.

(6)

4.

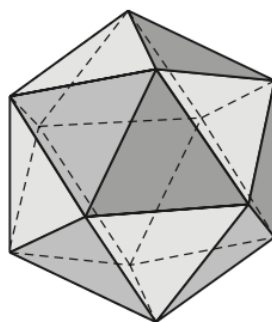


Figure 1

A regular icosahedron of side length x cm, shown in Figure 1, is expanding uniformly.

The icosahedron consists of 20 congruent equilateral triangular faces of side length x cm.

(a) Show that the surface area, A cm², of the icosahedron is given by

$$A = 5\sqrt{3}x^2 \quad (2)$$

Given that the volume, V cm³, of the icosahedron is given by

$$V = \frac{5}{12}(3 + \sqrt{5})x^3$$

(b) show that $\frac{dV}{dA} = \frac{(3 + \sqrt{5})x}{8\sqrt{3}} \quad (3)$

The surface area of the icosahedron is increasing at a constant rate of $0.025 \text{ cm}^2 \text{ s}^{-1}$

(c) Find the rate of change of the volume of the icosahedron when $x = 2$, giving your answer to 2 significant figures. (3)

3.

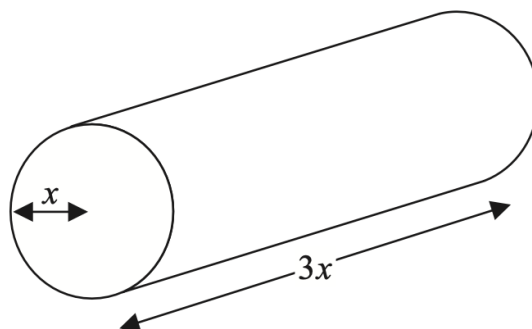


Figure 1

A tablet is dissolving in water.

The tablet is modelled as a cylinder, shown in Figure 1.

At t seconds after the tablet is dropped into the water, the radius of the tablet is x mm and the length of the tablet is $3x$ mm.

The cross-sectional area of the tablet is decreasing at a constant rate of $0.5 \text{ mm}^2 \text{ s}^{-1}$

(a) Find $\frac{dx}{dt}$ when $x = 7$

(4)

(b) Find, according to the model, the rate of decrease of the volume of the tablet when $x = 4$

(4)

2022 June Q4 P4

4. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve has equation

$$16x^3 - 9kx^2y + 8y^3 = 875$$

where k is a constant.

(a) Show that

$$\frac{dy}{dx} = \frac{6kxy - 16x^2}{8y^2 - 3kx^2}$$

(4)

Given that the curve has a turning point at $x = \frac{5}{2}$

(b) find the value of k

(4)

2022 October Q10 P4

10. A spherical ball of ice of radius 12 cm is placed in a bucket of water.

In a model of the situation,

- the ball remains spherical as it melts
- t minutes after the ball of ice is placed in the bucket, its radius is r cm
- the rate of decrease of the radius of the ball of ice is inversely proportional to the square of the radius
- the radius of the ball of ice is 6 cm after 15 minutes

Using the model and the information given,

(a) find an equation linking r and t , (5)

(b) find the time taken for the ball of ice to melt completely. (2)

(c) On Diagram 1 on page 27, sketch a graph of r against t . (1)

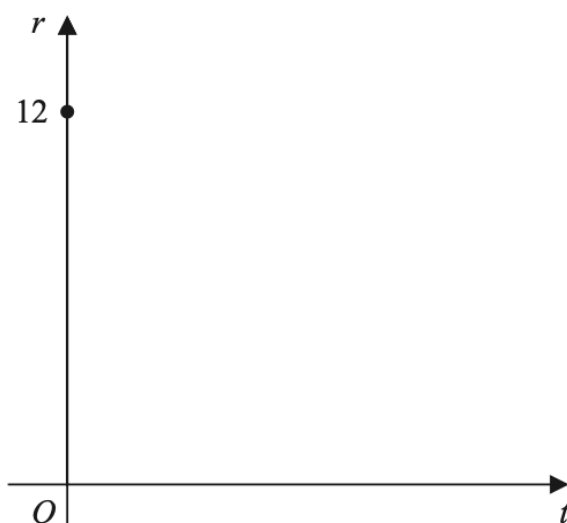


Diagram 1

11.

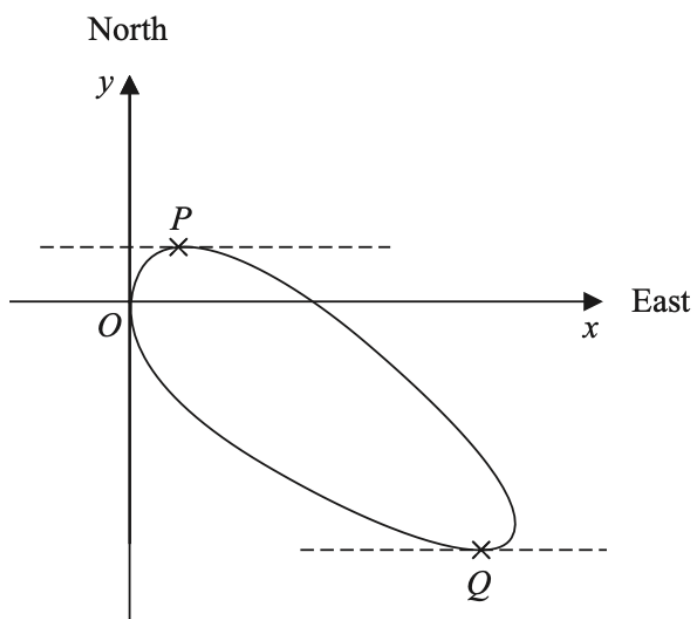


Figure 4

Figure 4 shows a sketch of the closed curve with equation

$$(x + y)^3 + 10y^2 = 108x$$

(a) Show that

$$\frac{dy}{dx} = \frac{108 - 3(x + y)^2}{20y + 3(x + y)^2}$$

(5)

The curve is used to model the shape of a cycle track with both x and y measured in km.

The points P and Q represent points that are furthest north and furthest south of the origin O , as shown in Figure 4.

Using the result given in part (a),

(b) find how far the point Q is south of O . Give your answer to the nearest 100 m.

(4)

5.

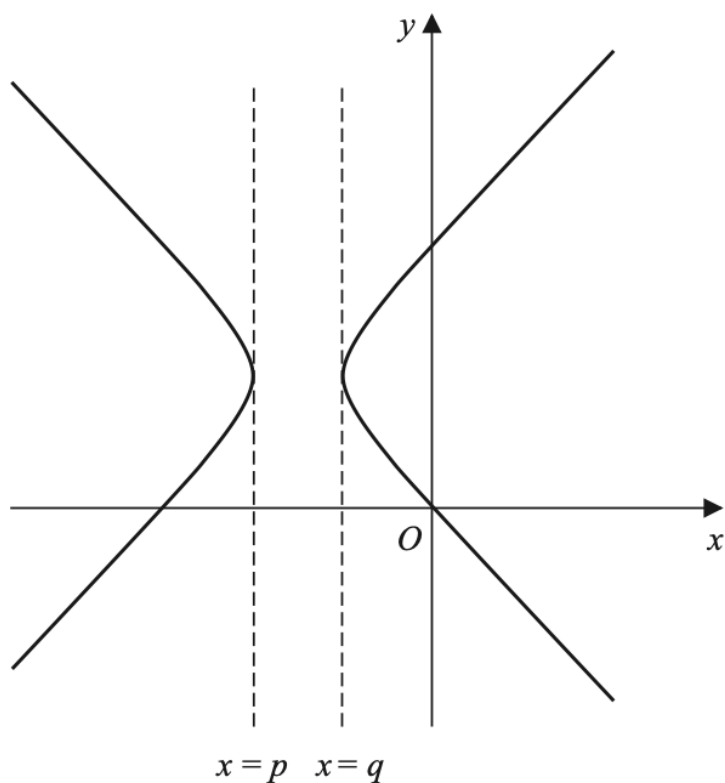


Figure 2

Figure 2 shows a sketch of the curve with equation

$$y^2 = 2x^2 + 15x + 10y$$

(a) Find $\frac{dy}{dx}$ in terms of x and y .

(4)

The curve is not defined for values of x in the interval (p, q) , as shown in Figure 2.

(b) Using your answer to part (a) or otherwise, find the value of p and the value of q .

(Solutions relying entirely on calculator technology are not acceptable.)

(3)

2023 January Q7 P4

7. The volume $V \text{ cm}^3$ of a spherical balloon with radius $r \text{ cm}$ is given by the formula

$$V = \frac{4}{3} \pi r^3$$

- (a) Find $\frac{dV}{dr}$ giving your answer in simplest form.

(1)

At time t seconds, the volume of the balloon is increasing according to the differential equation

$$\frac{dV}{dt} = \frac{900}{(2t + 3)^2} \quad t \geq 0$$

Given that $V = 0$ when $t = 0$

- (b) (i) solve this differential equation to show that

$$V = \frac{300t}{2t + 3}$$

- (ii) Hence find the upper limit to the volume of the balloon.

(5)

- (c) Find the radius of the balloon at $t = 3$, giving your answer in cm to 3 significant figures.

(3)

- (d) Find the rate of increase of the radius of the balloon at $t = 3$, giving your answer to 2 significant figures. Show your working and state the units of your answer.

(3)

2.

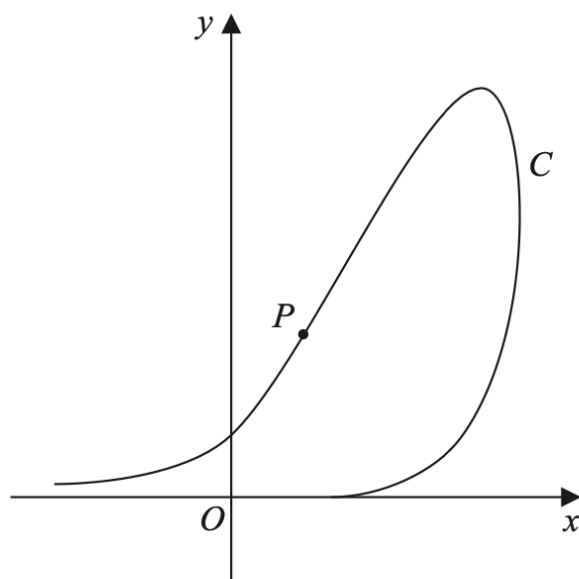


Figure 1

Figure 1 shows a sketch of part of the curve C with equation

$$2^x - 4xy + y^2 = 13 \quad y \geq 0$$

The point P lies on C and has x coordinate 2

(a) Find the y coordinate of P .

(2)

(b) Find $\frac{dy}{dx}$ in terms of x and y .

(5)

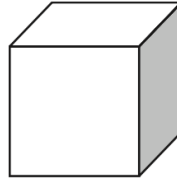
The tangent to C at P crosses the x -axis at the point Q .

(c) Find the x coordinate of Q , giving your answer in the form $\frac{a \ln 2 + b}{c \ln 2 + d}$ where a, b, c and d are integers to be found.

(3)

2023 October Q2 P4

2.



x cm

Figure 1

Figure 1 shows a cube which is increasing in size.

At time t seconds,

- the length of each edge of the cube is x cm
- the surface area of the cube is $S\text{cm}^2$
- the volume of the cube is $V\text{cm}^3$

Given that the surface area of the cube is increasing at a constant rate of $4\text{cm}^2\text{s}^{-1}$

(a) show that $\frac{dx}{dt} = \frac{k}{x}$ where k is a constant to be found, (4)

(b) show that $\frac{dV}{dt} = V^p$ where p is a constant to be found. (3)

5.

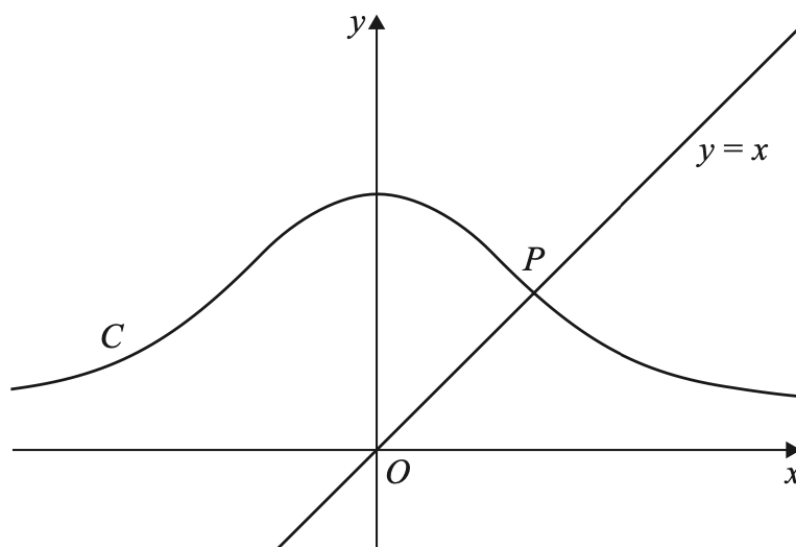


Figure 2

Figure 2 shows a sketch of the curve C with equation

$$y^3 - x^2 + 4x^2y = k$$

where k is a positive constant greater than 1

(a) Find $\frac{dy}{dx}$ in terms of x and y .

(5)

The point P lies on C .

Given that the normal to C at P has equation $y = x$, as shown in Figure 2,

(b) find the value of k .

(5)

3.

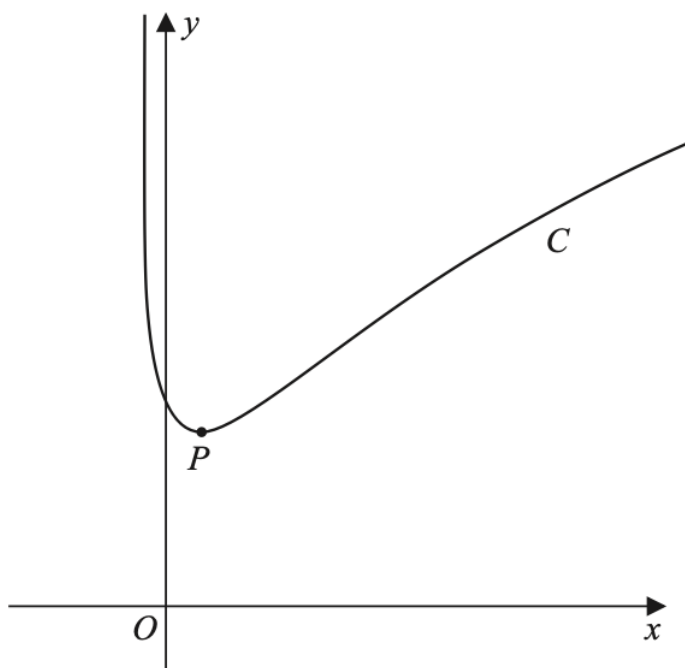


Figure 1

The curve C , shown in Figure 1, has equation

$$y^2x + 3y = 4x^2 + k \quad y > 0$$

where k is a constant.

(a) Find $\frac{dy}{dx}$ in terms of x and y

(5)

The point $P(p, 2)$, where p is a constant, lies on C .

Given that P is the minimum turning point on C ,

(b) find

(i) the value of p

(ii) the value of k

(4)

4.

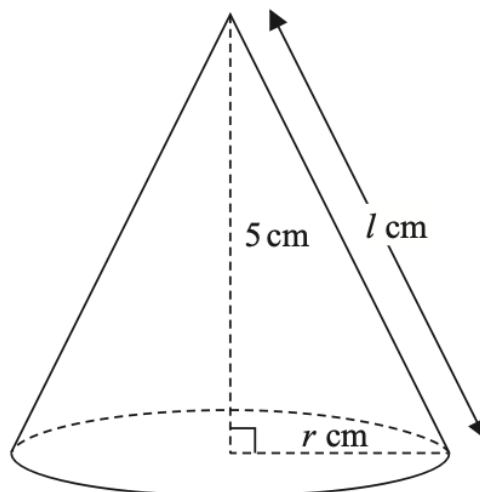


Figure 2

A cone, shown in Figure 2, has

- fixed height 5 cm
- base radius r cm
- slant height l cm

(a) Find an expression for l in terms of r

(1)

Given that the base radius is increasing at a constant rate of 3 cm per minute,

(b) find the rate at which the total surface area of the cone is changing when the radius of the cone is 1.5 cm. Give your answer in cm^2 per minute to one decimal place.

[The total surface area, S , of a cone is given by the formula $S = \pi r^2 + \pi r l$]

(4)